

Question	Marking Guidance	Mark	Comments
07.1	1. Same/stream water/gravel so natural/usual/normal habitat/environment OR Same/stream water/gravel so normal/usual behaviour OR Same/stream water/gravel so variables controlled/same OR Same/stream water so oxygen/pH controlled/same; 2. 30/large sample so representative/reliable; 3. (Placed at) centre so same distance to each segment/light intensity OR (Placed at) centre so no bias; 4. (10 minute) delay so shrimps can acclimatise/equilibrate; OR (10 minute) delay so shrimps enough time to respond/react/distribute OR (10 minute) delay so shrimps not stressed;	3 max (3 x AO3)	1. First 3 options can refer to same stream/water/gravel or stream water/gravel 1. Accept 'so light is the only variable' for 'so variables controlled' 2. Accept 30/large sample so reduces effect of anomalies 2. Ignore 'so reliable mean' obtained 4. Accept 'get used to the conditions' for acclimatise 4. Enough time to move 'without qualification' is insufficient for mark

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07.2	1. (Negative photo) taxis; 2. Predation less likely OR Less visible to predators/prey OR Less likely to be prey OR Move away from heat OR Move towards food;	2 (2 x AO2)	2. Accept 'consumers' for 'predators' 2. Accept 'stops predation' 2. Accept 'hides from predator/prey' 2. Accept descriptions of reduced predation e.g. 'fewer are eaten' 2. Accept 'move to cooler place /conditions'

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07.3	1. There is no (significant) difference in the distribution/ movement /behaviour/number of shrimps at different light intensities OR (Different) light intensity/intensities has/have no effect/relationship on/with the distribution/movement/behaviour of shrimp(s); 2. The data are discrete/categoric OR (To determine) if there is a difference between the observed and expected (results/data/values/frequencies);	2 (2 x AO2)	1. Accept 'shrimps show no preference for (different) light intensity/intensities 1. Accept correlation for mark point 1 but ignore for mark point 2.

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07.4	Route A – 4 max 1. (The calculated / χ^2 (value)/11.87) is greater than 7.82 / 11.35; 2. The probability of the difference being due to chance is less than 5% / 0.05 / 1% / 0.01 OR The probability of the difference not being due to chance is more than 95% / 0.95 / 99% / 0.99; 3. There is a significant difference; 4. Reject the null hypothesis; Route B – 2 max Ignore mark point 1 and 4 2. The probability of the difference being due to chance is more than 5% / 0.05 / 1% / 0.01 OR The probability of the difference not being due to chance is less than 95% / 0.95 / 99% / 0.99; 3. There is no significant difference;	4 (4 x AO3)	Note. All 4 marks can only be obtained via route A. There are 2 max marks for route B ignoring mark points 1 and 4. Route B allows credit for understanding of probability although context incorrect. Use the route providing most marks. 2. Reject 0.005% / 0.05% / 50% / 0.001% / 0.01% / 10% 2. Accept P for probability 2 and 3. if no reference to difference is the only error in both 2 and 3 allow one mark 4. Accept Ho for null hypothesis
